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Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2026

Code: 104606 ***Time: 03 Hours***

Semester: VI ***Subject: Digital CMOS VLSI Design*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) For an n-channel enhancement MOSFET, a conducting channel between source and drain is formed when

(i) $V_{GS} = 0$ (iii) $V_{GS} > V_{TH}$
(ii) $V_{GS} < 0$ (iv) $V_{DS} = 0$

(b) The primary purpose of photolithography in IC fabrication is

(i) Increase transistor current (iii) Measure transistor resistance
(ii) Transfer a geometric pattern onto the wafer (iv) Remove all oxide from the wafer

(c) Increasing the interconnect capacitance in a CMOS circuit generally

(i) Reduces propagation delay (iii) Increases threshold voltage
(ii) Eliminates static power (iv) Increases propagation delay

(d) For a simple lumped RC interconnect, the approximate 50%-transition delay is

(i) RC (iii) $0.69RC$
(ii) $0.5RC$ (iv) $2.2RC$

(e) The major advantage of CMOS logic over conventional NMOS logic is

(i) Higher static power consumption (iii) Zero transistor count
(ii) Very low static power dissipation (iv) Absence of complementary devices

(f) An edge-triggered flip-flop changes its stored state

(i) Only near a specified clock edge (iii) Throughout the entire clock HIGH period
(ii) Throughout the entire clock LOW period (iv) Only when both inputs are HIGH

(g) Which parameter is commonly used to represent the transconductance of a MOSFET?

(i) g_m (iii) C_{ox}
(ii) r_o (iv) V_{th}

(h) Layout design rules specify

(i) Only transistor threshold voltage (iii) Geometrical constraints between different layout layer
(ii) Only supply voltage (iv) Only transistor gain

(i) The dominant parasitic capacitance of a long on-chip interconnect is generally

(i) Wire-to-substrate and wire-to-wire capacitance (iii) Junction capacitance only
(ii) Oxide capacitance only (iv) Gate capacitance

(j) In an active-HIGH NOR-based SR latch, the condition S=R=1 is

(i) Set (iii) Hold
(ii) Reset (iv) Invalid/forbidden

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Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Explain the structure and operation of an n-channel enhancement MOSFET with a neat labeled diagram.

(b) An NMOS transistor has: V_{TH}=1V, k_n=2 mA/V^2, V_{GS}=3 V, V_{DS} = 4V. Determine: (i) The operating region and (ii) The drain current (I_D).

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Explain the CMOS n-Well Fabrication Process.

(b) Explain the design of a full-custom CMOS inverter layout.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain Switching Characteristics of a CMOS Inverter.

(b) Explain the operation of an NMOS resistive-load inverter with a neat circuit diagram.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Derive the propagation delay of a CMOS inverter using the equivalent RC model.

(b) A CMOS inverter has an effective pull-down resistance of 2 kΩ and drives a load capacitance of 50 fF.
(i) Calculate the propagation delay
(ii) If the maximum permissible delay is 100 ps, determine whether the design satisfies the requirement.
(iii) Calculate the maximum allowable load capacitance for a delay of 100 ps.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Design and explain a CMOS implementation of: $Y=\overline{AB + CD}$.

(b) Explain the structure and working of CMOS transmission gates.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Explain the difference between a latch and a flip-flop.

(b) Explain the operation of an edge-triggered D flip-flop.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Explain the current-voltage characteristics of an NMOS transistor.

(b) Explain the important verification techniques used after CMOS layout design.

Q.9 Write short notes on any two of the following: **[7 x 2 = 14]**

(a) Effect of transistor sizing on delay

(b) coupling capacitance

(c) Pass transistor

(d) Master slave configuration

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